

6. DIFFIE–HELLMAN KEY EXCHANGE

AIM

To implement the Diffie–Hellman Key Exchange algorithm using Python.

Algorithm

1. Start.
2. Choose a prime number and primitive root.
3. Enter Alice's private key.
4. Enter Bob's private key.
5. Calculate public keys.
6. Exchange public keys.
7. Calculate the shared secret.
8. Display the shared key.
9. Stop.

Python Code

```
prime = 23
primitive_root = 5
alice_private = int(input("Enter Alice's Private Key: "))
bob_private = int(input("Enter Bob's Private Key: "))
alice_public = (primitive_root ** alice_private) % prime
bob_public = (primitive_root ** bob_private) % prime
alice_secret = (bob_public ** alice_private) % prime
bob_secret = (alice_public ** bob_private) % prime
print("\nPrime Number :", prime)
print("Primitive Root :", primitive_root)
print("\nAlice Public Key :", alice_public)
print("Bob Public Key   :", bob_public)
print("\nAlice Shared Secret :", alice_secret)
print("Bob Shared Secret   :", bob_secret)
```

Output

```
... Enter Alice's Private Key: 6
Enter Bob's Private Key: 15

Prime Number : 23
Primitive Root : 5

Alice Public Key : 8
Bob Public Key : 19

Alice Shared Secret : 2
Bob Shared Secret : 2
```

The result:

The above code is executed and the output is obtained successfully.